

Periodic Table

1 H 1.008	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
3 Li 6.941	4 Be 9.012	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	11 Na 22.99	12 Mg 24.31	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95	19 K 39.10	20 Ca 40.08
21 Sc	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.59	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	37 Rb 85.47	38 Sr 87.62
39 Y	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	55 Cs 132.9	56 Ba 137.3
71 La	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po	85 At	86 Rn	87 Fr	88 Ra
89 Ac	90 Th 232.0	91 Pa	92 U 238.0	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	104 Rf	105 Db	106 Sg
107 Bh	108 Hs	109 Mt	110 Ds	111 Nh	112 Fl	113 Mc	114 Lv	115 Ts	116 Og	117 Nh	118 Fl	119 Mc	120 Lv	121 Ts	122 Og	123 Nh	124 Fl

6 C 12.01	Atomic Number	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
Atomic Mass	140.9	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0	176.3
Symbol	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es
Series	Actinide Series	Lanthanide Series	Actinide Series	Lanthanide Series	Actinide Series	Lanthanide Series	Actinide Series	Lanthanide Series	Actinide Series	Lanthanide Series

Formulae

Number of moles	$n = \frac{m}{M}$	mass	molar mass
Number of moles of solute	$n = \frac{cV}{22.41}$	volume	concentration
Number of moles of a gas at STP	$n = \frac{V}{22.41}$	volume	concentration
Ideal gas law	$PV = nRT$	pressure	volume
Parts per million	$\text{ppm} = \frac{\text{mass of solute (mg)}}{\text{mass of solution (kg)}}$	mass of solute	mass of solution
pH of a solution	$\text{pH} = -\log [H^+]$	concentration	pH
Density	$\rho = \frac{\text{mass of sample}}{\text{volume of sample}}$	mass of sample	volume of sample

Units

Volumes are given in the units of litres (L), or millilitres (mL).
Temperatures are given in the units of degrees Celsius (°C) or Kelvins (K).
It may be assumed that 0.0°C = 273.1 K.
Energy changes are given in the SI unit kilojoule (kJ).
Pressures are given in the SI unit kilopascal (kPa) and in atmospheres (atm).
S.T.P. is 0.0°C and 101.3 kPa.
Solution concentrations are given in the units moles per litre (mol L⁻¹), grams per litre (g L⁻¹) and parts per million.
Constants
Universal gas constant, R = 8.315 J K⁻¹ mol⁻¹
Avogadro constant, N = 6.022 x 10²³ mol⁻¹
Volume of 1.000 mol of an ideal gas at 0.0°C and 101.3 kPa is 22.41 L
S.T.P. is 0.0°C and 101.3 kPa
Equilibrium constant for water at 25°C, K_w = 1.00 x 10⁻¹⁴

Solubility rules for ionic solids in water

Soluble	Insoluble	Exceptions	Slightly soluble
Most chlorides	AgCl	PbCl ₂	
Most bromides	AgBr	PbBr ₂	
Most iodides	AgI, PbI ₂		
All nitrates		No exceptions	
All ethanoates			
Most sulfates	SrSO ₄ , BaSO ₄ , PbSO ₄	CaSO ₄ , Ag ₂ SO ₄	

Insoluble in water

Insoluble	Soluble	Exceptions	Slightly soluble
Most hydroxides	NaOH, KOH, Ba(OH) ₂ (note: NH ₄ OH and AgOH do not exist)	Ca(OH) ₂ , Sr(OH) ₂	
Most carbonates	Na ₂ CO ₃ , K ₂ CO ₃ , (NH ₄) ₂ CO ₃		
Most phosphates	Na ₃ PO ₄ , K ₃ PO ₄ , (NH ₄) ₃ PO ₄		
Most sulfides	Na ₂ S, K ₂ S, (NH ₄) ₂ S		

Soluble = more than 0.1 mole dissolves per litre
Slightly soluble = between 0.01 and 0.1 mole dissolves per litre
Insoluble = less than 0.01 mole dissolves per litre

Coloured ions in aqueous solution

Cation	Colour
Cr ³⁺	deep green
Co ²⁺	pink
Cu ²⁺	blue
Fe ²⁺	pale green
Fe ³⁺	brown
Mn ²⁺	very pale pink
Ni ²⁺	green
Anion	Colour
CrO ₄ ²⁻	yellow
Cr ₂ O ₇ ²⁻	orange
MnO ₄ ⁻	purple

Coloured halogens

Halogen	Colour of free element
F ₂ (g)	yellow
Cl ₂ (g)	green
Br ₂ (l)	red
I ₂ (s)	dark grey
Halogen	Colour of halogen in aqueous solution
Cl ₂ (aq)	pale yellow
Br ₂ (aq)	orange
I ₂ (aq)	brown
Halogen	Colour of halogen in organic solvent
Br ₂	red
I ₂	purple